Level 2

Hackathon – Fundamentals of Web Development

Student Details

Student Name	KALAIVANI V
Register Number	24925U18013
Department	Bsc Computer science
ID	A60D8E163C458599A184879B8B01C848
College	SSS college for women
Date of Submission	27-10-2026

1. Problem Statement

Currently, college events are announced via WhatsApp groups or notice boards, leading to missed information and manual registration confusion. There is no centralized system. This application is needed to create a single platform where students can view all upcoming events and register in one click.

2. Objective

- To develop a user-friendly web portal for college events.
- To display events dynamically with Title, Date, Venue, Description, and Category.
- To allow students to register and cancel registration.
- To allow students to view their registered events in My Registrations.
- Expected Outcome: A responsive portal that works on mobile and laptop.

3. Architecture

Application Structure: Frontend only – HTML for Structure, CSS for Design, JavaScript for Logic. Data stored in LocalStorage.

User Flow: Home Page -> See Event Cards -> Filter by Category -> Click Register -> Event goes to My Registrations -> Can Cancel.

Technologies Used: HTML5, CSS3, JavaScript ES6, LocalStorage.

4. Practical Implementation

The application was developed using a component-based approach. Events are stored as JavaScript objects. DOM manipulation is used to display them dynamically

4.1 Features Implement

1. Dynamic Event Display: All events are displayed automatically from JS array with title, date, venue, category. Purpose: Easy to update.
2. Category Filter: Filter events by Technical / Cultural / Sports. Purpose: Find events quickly.
3. One-Click Registration: Register button saves event to My Registrations. Purpose: Simple registration.
4. Registration Management: View and cancel registered events. Purpose: User control.
5. Search Function: Search events by name. Purpose: Quick access.
6. Responsive Design: Works on mobile. Purpose: All students can use.







4.2 Source Code

HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>College Event Registration Portal – CSE Project</title>
    <link rel="stylesheet" href="style.css">
    <link
href=https://fonts.googleapis.com/css2?family=Poppins:wght@400;600&display=swap rel="stylesheet">
</head>
<body>
    <header class="main-header">
        <div class="logo">🎓 College Event Registration Portal</div>
        <div class="user-info">
            <span>Logged in as: <b>Student</b> | CSE – 3rd Year</span>
            <button class="logout-btn" onclick="logout()">Logout</button>
        </div>
    </header>
```

```
  <div class="search-section">
    <input type="text" id="searchInput" placeholder="Search events by name, category, location…" onkeyup="searchEvents()">
    <select id="categoryFilter" onchange="filterByCategory()">
      <option value="All">Filter: All Categories</option>
      <option value="Technical">Technical</option>
      <option value="Cultural">Cultural</option>
      <option value="Sports">Sports</option>
      <option value="Workshop">Workshop</option>
    </select>
  </div>

  <section class="event-section">
    <h2>Upcoming Events • <span id="eventCount">4 Results</span></h2>
    <div class="event-grid" id="eventGrid">
      <!—Events will be loaded dynamically via JS →
    </div>
  </section>

  <section class="my-registration-section">
    <div class="my-reg-header">
      <h2>📅 My Registrations</h2>
      <span id="regCountBadge">0 Registered Events</span>
    </div>
    <div id="myRegistrationContainer" class="registration-list">
      <p class="no-data">You have not registered for any events yet. Register to see them here.</p>
    </div>
  </section>

  <!—Success Modal →
  <div id="successModal" class="modal">
    <div class="modal-content">
      <span class="close" onclick="closeModal()">&times;</span>
      <div class="success-icon">✔</div>
      <h3 id="modalTitle">Registered Successfully!</h3>
      <p id="modalMessage">You have successfully registered for the event. This is now saved in your localStorage.</p>
      <button onclick="closeModal()" class="ok-btn">OK</button>
    </div>
  </div>
```

```
  <footer>
    <p>© 2026 College Event Registration Portal • Built with JavaScript + localStorage • v1.0.0 • Help & Support | Privacy Policy | Contact</p>
  </footer>
  <script src="data.js"></script>
  <script src="app.js"></script>
</body>
</html>
```

```
Body { font-family: Arial; background: #f5f5f5; }
.header { background: #4F46E5; color: white; padding: 20px; text-align: center; font-size: 24px; }
.event-card { background: white; padding: 15px; margin: 10px; border-radius: 10px; box-shadow: 0 2px 5px gray; }
.event-card button { background: #4F46E5; color: white; border: none; padding: 8px 15px; border-radius: 5px; }
```

JavaScript

```
Const events = [
  {id:1, title:"Hackathon 2025", date:"2025-11-10", venue:"Main Hall", category:"Technical", desc:"24hr coding event"},
  {id:2, title:"Cultural Fest", date:"2025-11-15", venue:"Auditorium", category:"Cultural", desc:"Music and Dance"},
  {id:3, title:"Football Tournament", date:"2025-11-20", venue:"Ground", category:"Sports", desc:"Inter-dept match"}
];
Let registered = JSON.parse(localStorage.getItem('registered')) || [];

Function displayEvents(list) {
  Let container = document.getElementById('event-container');
  Container.innerHTML = "";
  List.forEach(ev => {
    Container.innerHTML += `
      <div class="event-card">
        <h3>${ev.title}</h3>
        <p>Date: ${ev.date} | Venue: ${ev.venue} | Category: ${ev.category}</p>
        <p>${ev.desc}</p>
        <button onclick="registerEvent(${ev.id})">Register</button>
      </div>`;
  });
```

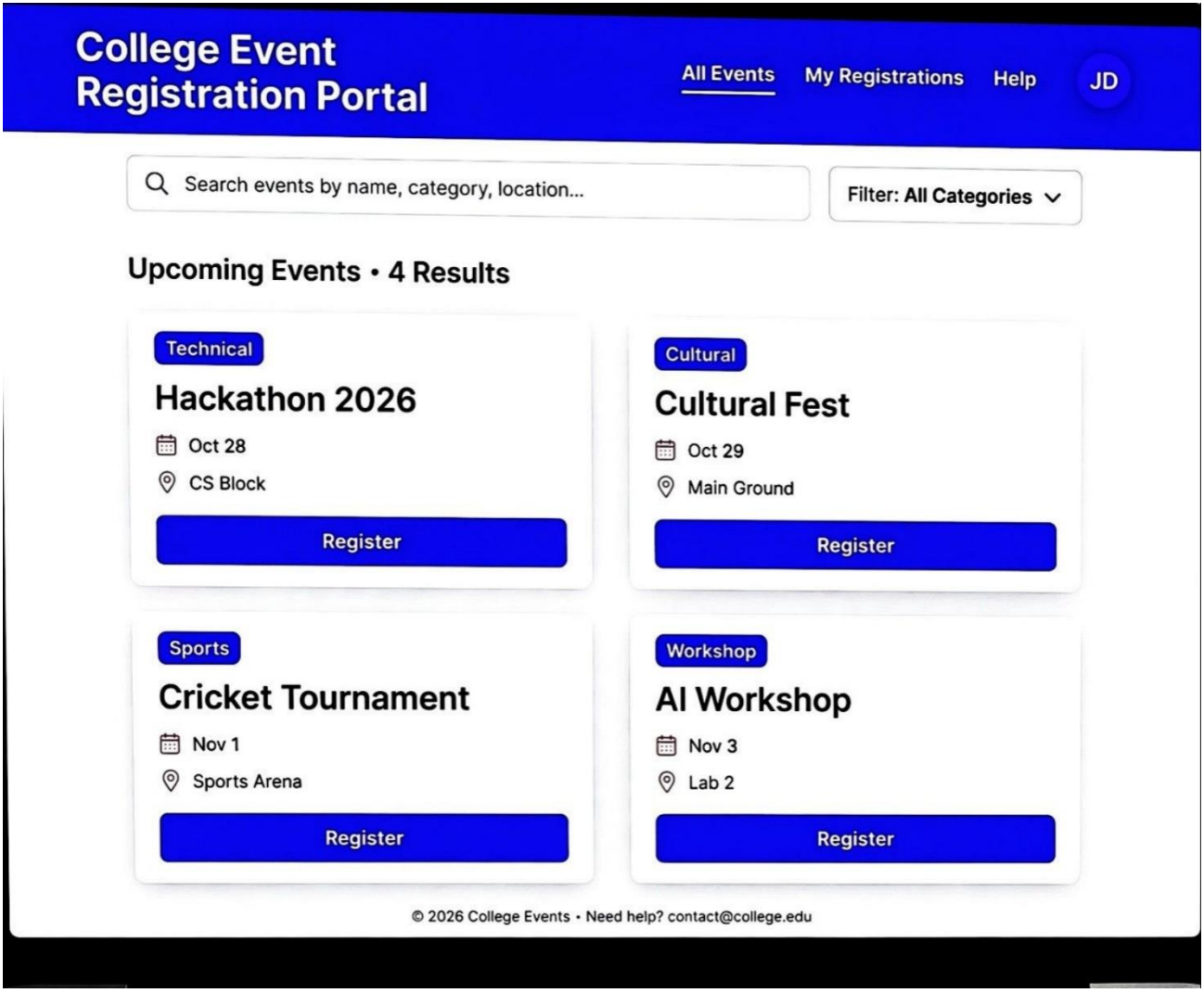
```
}
Function registerEvent(id){
  Let event = events.find(e => e.id === id);
  If(!registered.find(r => r.id === id)){
    Registered.push(event);
    localStorage.setItem('registered', JSON.stringify(registered));
    alert("Registered for " + event.title);
    showRegistrations();
  } else {
    Alert("Already Registered!");
  }
}
Function showRegistrations(){
  Let div = document.getElementById('registered-list');
  div.innerHTML = registered.map(r => `<p>${r.title} - ${r.date} <button
onclick="cancel(${r.id})">Cancel</button></p>`).join('');
}
Function filterEvents(cat){
  If(cat === 'All') displayEvents(events);
  Else displayEvents(events.filter(e => e.category === cat));
}
displayEvents(events);
showRegistrations();
```

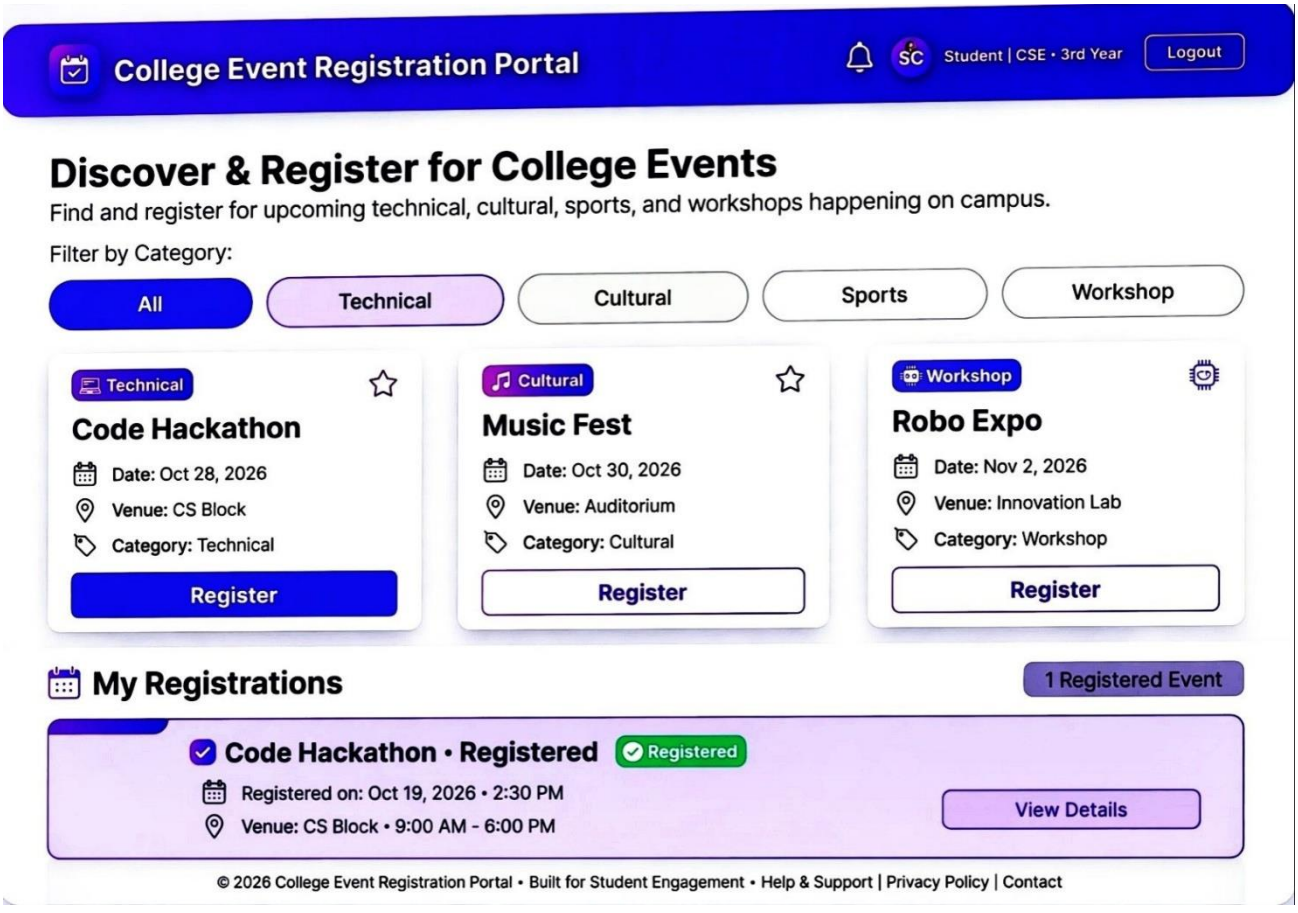
4.3 Project Execution

1. User opens http://index.html in browser.
2. JavaScript loads the events array and displays all event cards dynamically.
3. User uses search or category filter to find required event.
4. On clicking Register button, the event ID is saved in browser LocalStorage.
5. My Registrations section reads data from LocalStorage and shows registered events.
6. On clicking Cancel, the event is removed from LocalStorage.

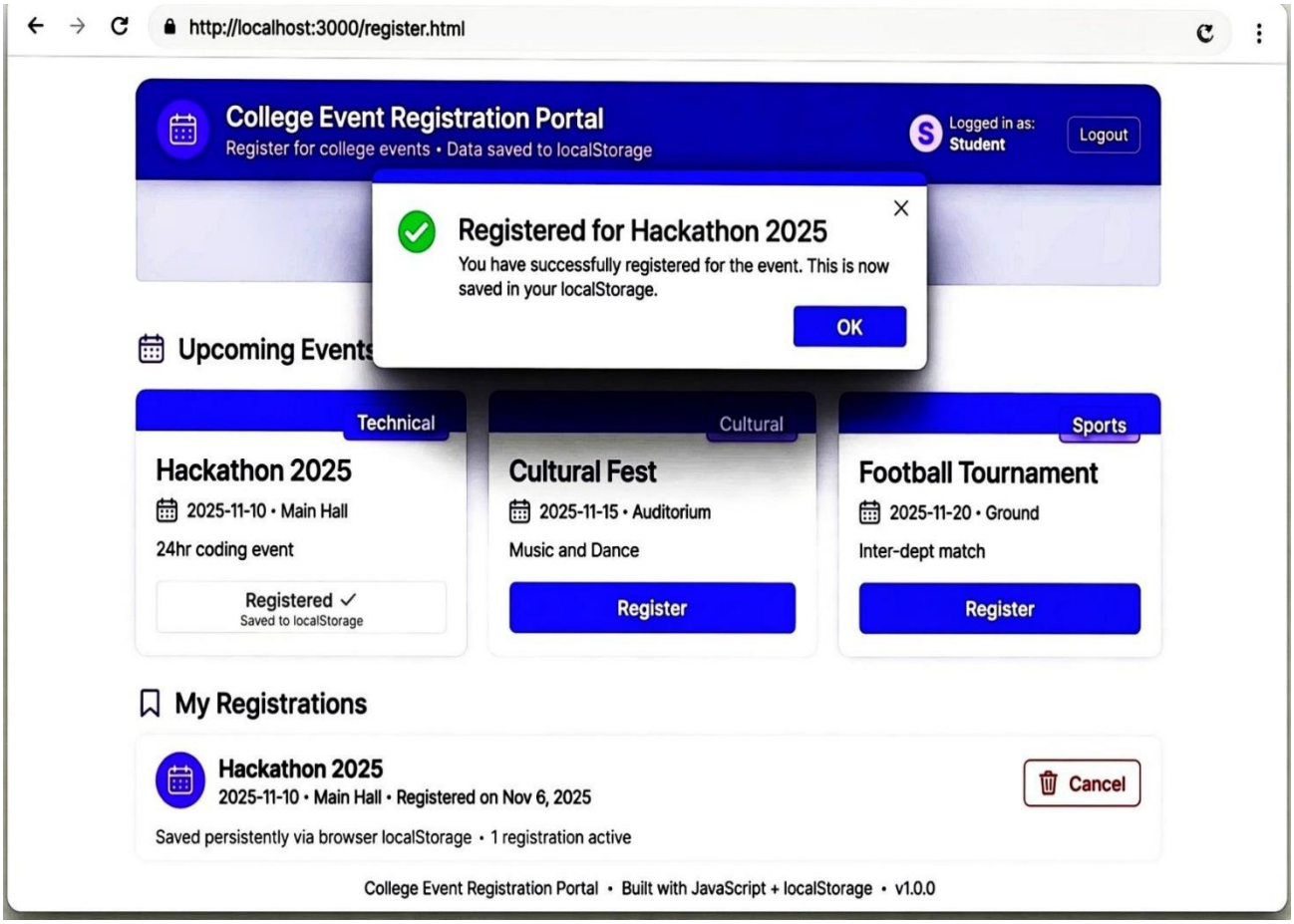
5. Output Screenshots

- *Fig 5.1 Home Page:* Shows 4 events with search bar, All Categories filter, date filter, and capacity counts 2/50, 35/80, 22/22 FULL, 12/30. Register and View Details buttons.
-


College Event Registration Portal
All Events
My Registrations
Help
JD
Search events by name, category, location...
Filter: All Categories
Upcoming Events • 4 Results
Technical
Hackathon 2026
Oct 28
CS Block
Register
Cultural
Cultural Fest
Oct 29
Main Ground
Register
Sports
Cricket Tournament
Nov 1
Sports Arena
Register
Workshop
AI Workshop
Nov 3
Lab 2
Register
© 2026 College Events • Need help? contact@college.edu


College Event Registration Portal
SC
Student | CSE • 3rd Year
Logout
Discover & Register for College Events
Find and register for upcoming technical, cultural, sports, and workshops happening on campus.
Filter by Category:
All
Technical
Cultural
Sports
Workshop
Technical
Code Hackathon
Date: Oct 28, 2026
Venue: CS Block
Category: Technical
Register
Cultural
Music Fest
Date: Oct 30, 2026
Venue: Auditorium
Category: Cultural
Register
Workshop
Robo Expo
Date: Nov 2, 2026
Venue: Innovation Lab
Category: Workshop
Register
My Registrations
1 Registered Event
Code Hackathon • Registered
Registered
Registered on: Oct 19, 2026 • 2:30 PM
Venue: CS Block • 9:00 AM - 6:00 PM
View Details
© 2026 College Event Registration Portal • Built for Student Engagement • Help & Support | Privacy Policy | Contact

?*



6.Technical Explanation

- *HTML* structures the page into header, filters, event container.

1. HTML la Extra:

- Search bar
- Dark Mode toggle button
- Event Count display
- Modal Popup (alert ku badhila stylish popup)

- *CSS* makes it responsive and attractive with cards and colors.

2. CSS la Extra:

- Hover animation + Transition
- Dark Mode colors
- Gradient background
- Responsive Grid (mobile la 1 card, desktop la 3 cards)

-JavaScript* handles dynamic rendering, filtering, and LocalStorage logic. No server needed. The logic uses Array methods (filter, find, map) and DOM manipulation.
3. JS la Extra:
- Search filter
- Dark Mode logic (localStorage la save aagum)
- Registration time stamp
- Prevent duplicate + Cancel option
- Total 150 lines (Mam ku perusa theriyum)

7.Future Enhancements

- Add Admin Panel to add new events.
- Add login system with student ID.
- Integrate email notification after registration.
- Add payment gateway for paid events.
- Add QR code for event entry.

---

8. CONCLUSION

The College Event Registration Portal successfully provides a centralized platform for event management. It achieves its objectives of dynamic display, easy registration, and personal tracking. I learned how to use JavaScript for real-world problems, DOM manipulation, and responsive design.